

YANSHUO CHEN

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RESEARCH INTERESTS

AI for biology: Generative modeling for virtual cells, mRNA design, and protein design.

Efficient machine learning: Mixed-precision post-training quantization, quantization-aware training, and pruning for language and multimodal models.

INDUSTRY EXPERIENCE

Research Intern, Adobe Research

May – Aug. 2026

Efficient Multi-Agent Workflow.

SERVICE & HONORS

Gold Reviewer, ICML

2026

Gold Medal, China National Biology Olympiad

Aug. 2017

SELECTED PUBLICATIONS

* Equal contribution.

MPQ: A Message-Passing View of Post-Training Quantization

Yanshuo Chen, Chenxi Liu, Reza Shirkavand, and Heng Huang.

NeurIPS 2026. Accepted as a poster.

Grounding Multimodal Reasoning with Evidence-Ablated Negatives

Chenxi Liu, Tianyi Xiong, Yanshuo Chen, and Heng Huang.

NeurIPS 2026. Accepted as a poster.

Multi-Crit: Benchmarking Multimodal Judges on Pluralistic Criteria-Following

Tianyi Xiong, Yi Ge, Ming Li, Zuolong Zhang, Pranav Kulkarni, Kaishen Wang, Qi He, Zeying Zhu, Chenxi Liu, Ruibo Chen, Tong Zheng,

Yanshuo Chen, Xiyao Wang, Renrui Zhang, Wenhui Chen, and Heng Huang.

CVPR 2026, Highlight. [\[paper\]](#) [\[project\]](#) [\[code\]](#)

Learning from Self-Debate: Preparing Reasoning Models for Multi-Agent Debate

Chenxi Liu, Yanshuo Chen, Ruibo Chen, Tianyi Xiong, Tong Zheng, and Heng Huang.

arXiv preprint, 2026. [\[paper\]](#) [\[code\]](#)

A Watermark for Order-Agnostic Language Models

Ruibo Chen*, Yihan Wu*, Yanshuo Chen, Chenxi Liu, Junfeng Guo, and Heng Huang.

ICLR 2025. [\[paper\]](#) [\[arXiv\]](#)

CodonTranslator: A Conditional Codon Language Model for Codon Optimization Across Life Domains

Yanshuo Chen*, Yuming Zhang*, Joshua Li, Boxue Tian, and Heng Huang.

bioRxiv preprint, 2025. [\[paper\]](#) [\[code\]](#) [\[model\]](#)

Fast and Scalable Wasserstein-1 Neural Optimal Transport Solver for Single-Cell Perturbation Prediction

Yanshuo Chen, Zhengmian Hu, Wei Chen, and Heng Huang.

Bioinformatics; ISMB/ECCB 2025. [\[paper\]](#) [\[arXiv\]](#) [\[code\]](#)

Enhancing Privacy in Biosecurity with Watermarked Protein Design

Yanshuo Chen*, Zhengmian Hu*, Yihan Wu, Ruibo Chen, Yongrui Jin, Marcus Zhan, Chengjin Xie, Wei Chen, and Heng Huang.

Bioinformatics, 41(7), btaf141, 2025. [\[paper\]](#) [\[code\]](#)

Deep Autoencoder for Interpretable Tissue-Adaptive Deconvolution and Cell-Type-Specific Gene Analysis

Yanshuo Chen*, Yixuan Wang*, Yuelong Chen, Yuqi Cheng, Yumeng Wei, Yunxiang Li, Jiuming Wang, Yingying Wei, Ting-Fung Chan, and Yu Li.

Nature Communications, 13, 6735, 2022. [\[paper\]](#) [\[code\]](#) [\[docs\]](#)